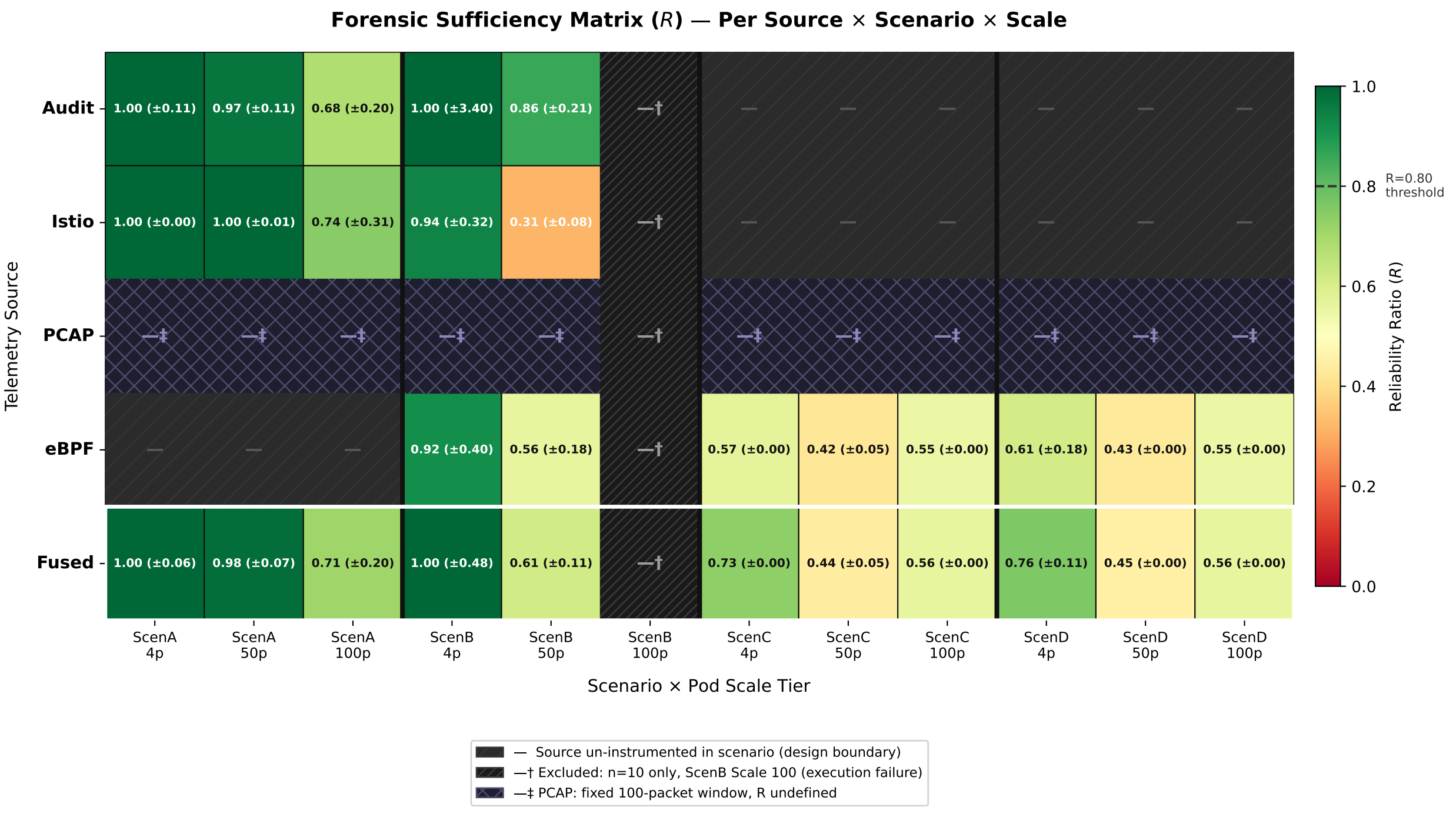
Forensic Sufficiency Matrix (R) — Per Source × Scenario × Scale
Telemetry Source
Audit
Istio
PCAP
eBPF
Fused
1.00 (±0.11)
0.97 (±0.11)
0.68 (±0.20)
1.00 (±3.40)
0.86 (±0.21)
1.00 (±0.00)
1.00 (±0.01)
0.74 (±0.31)
0.94 (±0.32)
0.31 (±0.08)
0.92 (±0.40)
0.56 (±0.18)
0.57 (±0.00)
0.42 (±0.05)
0.55 (±0.00)
0.61 (±0.18)
0.43 (±0.00)
0.55 (±0.00)
1.00 (±0.06)
0.98 (±0.07)
0.71 (±0.20)
1.00 (±0.48)
0.61 (±0.11)
0.73 (±0.00)
0.44 (±0.05)
0.56 (±0.00)
0.76 (±0.11)
0.45 (±0.00)
0.56 (±0.00)
—†
—‡
—
ScenA 4p
ScenA 50p
ScenA 100p
ScenB 4p
ScenB 50p
ScenB 100p
ScenC 4p
ScenC 50p
ScenC 100p
ScenD 4p
ScenD 50p
ScenD 100p
Scenario × Pod Scale Tier
Reliability Ratio (R)
1.0
0.8
0.6
0.4
0.2
0.0
R=0.80 threshold
— Source un-instrumented in scenario (design boundary)
—† Excluded: n=10 only, ScenB Scale 100 (execution failure)
—‡ PCAP: fixed 100-packet window, R undefined